School of Computing and Creative Technologies

Assessment Brief

Submission details

Module title:	Machine Learning
Module code:	UFCFAS-15-2
Module Leader:	Professor Jun Hong
Assessment title:	Group Project
Year:	2025-26
Assessment type:	Group Project
Assessment weighting:	70% of total module mark
Size or length of assessment:	Project proposal (10%): One page Project Report (60%): Six pages Please refer to the Formatting Guidelines and Word template in the corresponding sections
Module learning outcomes assessed by this task:	MO2 Select and apply machine learning algorithms to formulate solutions to different types of machine learning problems, taking into account criteria such data availability and characteristics, and problem-specific requirements for balancing speed, accuracy, and explainability. MO3 Implement and evaluate contemporary machine learning solutions to application problems using a range of contemporary frameworks. MO4 Demonstrate an awareness of the ethical and societal implications of machine learning solutions.
Use of AI in assessment:	You are permitted to use Gen AI tools in an assistive role in this assessment to support you with: • Researching your real-world application problem • Finding suitable sources of information on related topics Please note that while GenAI tools might suggest information/content, you will always need to go back and locate and critically engage with actual literature/research/evidence/information yourself, and make sure to show that you create and fully understand all the content yourself in your report.
Guidance on Referencing (inc AI):	Please note that the aim of referencing is to demonstrate you have read and understood a range of sources to evidence your key points. You need to list the references consistently and in such a way as to ensure the reader can follow up on the sources for themselves. Please refer to the Formatting Guidelines section for the required referencing style.

Submission and feedback dates

Submission deadlines:	Before 17:00 on 13 March 2026 for the project proposal. Before 17:00 on 1 May 2026 for the project final submission. Both are eligible for 48-hour late submission window. All times are 24-hour clock, current local time (at time of submission) in the UK. Please refer to the project schedule for further information.
Submission format:	Please refer to the Formatting Guidelines section for further information.
Marks and Feedback due on:	They will be released no later than 20 working days after the late submission window date.
Marks and Feedback will be provided via:	They will be provided on Blackboard. Please refer to the Marking and Feedback section for further information.

Group Project (Assessment Task B) Specification (2025/26)

Each group typically consists of three students, working on a project under the supervision of a module tutor. Your project will be to design, develop, evaluate and present an end-to-end ML solution in Python to a real-world application problem. There are two deliverables in the project, *a project proposal and a final submission*. The project proposal is one page with a specified structure. The final submission consists of a research paper-like project report with a specified structure plus the Python code (i.e. Jupyter notebook with the appropriate markdown and code cells) of the project.

Assignment Schedule

The group project specification is **handed out in Week 3** (starting 9 Febuary 2026).

The group allocation: starts in Week 3 with an allocation deadline: 5pm, Friday in Week 5, 27 Febuary 2026.

The project proposals (10% of the total module mark) are due: 5pm, Friday in Week 7, 13 March 2026 (the grace period of 48 hours applies).

Submission: Please submit your proposal as a PDF on Blackboard. *Only one person on your group should submit.* Please list your group ID and group members (student numbers and names, with each named member indicating the participation of the member in the submission) on the submission pop-up page. Being named in the submission indicates the

The project final submissions (60% of the total module mark) are due: 5pm, Friday in Week 12, 1 May 2026 (the grace period of 48 hours applies).

Total Group Project Weighting: 70% of the total module mark.

Submission: Please submit your final project report as a PDF and your supplementary material as a separate ZIP file on Blackboard.

Please note that any text (e.g. in markdown cells if a Jupyter Notebook is used or comments in the code) in the suppelemntary material is for the coding and documentation purposes only and does not directly contribute the marking of the project report. No use of extensive texts is expected.

Only one person on your group should submit. *Please list your group ID and group members (student numbers and names, with each named*

member indicating the participation of the member in the submission) on the submission pop-up page.

The submission links for both the proposal and final submission are in the assessment folder on the module site on the Blackboard

Project Proposal

The first step is to work together as a group to write a project proposal. This may take several rounds of revision to become something that everyone is happy with, including your module tutor.

The project proposal should be one page (up to 400 words). It should describe:

- *The application problem*
  What is the application problem that you will be investigating? Why is it important?
- *Context and background*
  A brief introduction to the project, the aims and objectives of the project, and what have you read or researched that led to this project?
- *The dataset*
  What data will you use? If you are collecting a new dataset, how will you do it? If it is an existing dataset, what is it? Why do you choose it?
- *Methodology*
  What algorithms are you proposing to apply? If there are existing implementations, will you use them and how? How do you plan to improve or modify such implementations? You don't have to have an exact answer at this point, but you should have a general sense of how you will approach the problem you are working on.
- *Evaluation*
  How will you evaluate your results? Qualitatively, what kind of results do you expect (e.g. plots or figures)? Quantitatively, what kind of analysis will you use to evaluate and/or compare your results (e.g. what performance metrics)?

Ensure your proposal is double column, 10pt font size, and includes the names of all authors (i.e., everyone in your project group).

Choosing a Real-world Problem

If you are having trouble thinking of ideas, are there any problems in the world that you care about that might benefit from a Machine Learning solution, i.e. for social good, making the world a better place? Is there something you really enjoy doing that would be fun to involve Machine Learning in, i.e. a fun problem?

These are just suggestions which you can ignore, and you are encouraged to choose a completely different category of problem.

Choosing/Creating A Dataset

There exist many datasets that can found online, e.g. on UCI Machine Learning Repository (https://archive.ics.uci.edu/), Kaggle (https://www.kaggle.com/datasets), and many sites that contain image datasets, e.g. Google Open Images Dataset (https://opensource.google/). These datasets have been selected for Machine Learning.

There also exist many real-world datasets that have been created for various purposes, e.g. Open Data Bristol (https://opendata.bristol.gov.uk/) contains a lot of datasets most of which are created by Bristol City Council. Indeed, there are many, many datasets for us to explore, which itself can be a good exploratory and discovery process.

Project Final Report

Your project final report is required to be *up to 6 pages using the provided template*, structured like a paper from an AI/Machine Learning conference.

The following is a suggested structure for your report, as well as the rubric that is used for marking your report:

Title, Author(s)

Abstract Briefly describe your application problem, approach, and key results. Should be no more than 200 words.

Introduction (10%) Describe the problem you are working on, the aims and objectives of the project, your motivations (i.e., why it's important), and an overview of your approach and results

Related Work (10%) Describe and critically analyse published work that relates to your project. How is the cited work related to your proposed solution? How is your approach similar or different from the related work?

Data (10%) Describe the data you are working with for your project. What type of data is it? Where did it come from? How much data are you working with? What are the key characteristics of the data? Did you do any pre-processing and other kinds of treatment to use this data in your project?

Methods (30%) Describe and discuss your approach/algorithms for solving the problem that you set up in the introduction. Why are your approach/algorithms the right thing to do? Did you consider alternative approaches/algorithms? It may be helpful to include figures, diagrams, or tables to describe your method or compare it with other methods.

Experiments (30%) Describe and siscuss the experiments that you performed to demonstrate that your approach solves the problem. The exact experiments will vary depending on the project, but you might compare with previously published methods, experiment with different hyperparameters or architectural choices, use appropriate performance metrics or visualization techniques to gain insights into how well your model works, discuss any limitations or weaknesses. You should include graphs, tables, or other figures to illustrate your experimental results.

Conclusion (5%) Summarize your key results – What have you achieved? What have you learned? Suggest ideas for future extensions or new applications of your ideas.

Writing and Formatting (5%) Is your paper clearly written and nicely formatted?

Formatting Guidelines

Refer to the appropriate Word template and PDF files in the Formatting Guidelines folder in the Assessment folder for the guidelines including the main body and referendces of the report.

There is no specific word count limit on the report. However, the formatting guidelines should be followed, including the 6-page limit and format, and the font size. Please note that the line numbers are not part of the report.

Expected amount of time individually: 80 hours.

Group Foramtion and Group Tutor Allocation

The group allocation will start on the voluntary basis in Week 3 when you are free to form your own group. Once your group is formed, please let a module tutor know, who will enrol your group. Any student who is yet to be in an enrolled group by Week 5 will be randomly allocated into a group.

Each project is expected to be allocated to a module tutor who will have a brief catch up meeting with the group during a lab session every two weeks so that the tutor can be informed of the group's progress and engagement.

Marking and Feedback

Your group project proposal will be marked with a grade of A, B, C, D, F only with a short paragraph of summary comments. More oral feedback may be provided by your allocated module tutor.

A holistic approach will be taken in marking the submission. Typically, the same group mark will be given to each individual member of the group if all the group members have engaged in the project as normally expected. However, an individual member's lack of engagement may result in the individual mark reduced from the group mark. An individual member's complete lack of engagement may result in a 0 mark.

Formative feedback will be provided by module tutors in lab sessions and at catch-up meetings. Summative feedback will be provided by the marking grid and comments in the completed group project assessment form as specified in the marking criteria. The group assessment form will be released within 4 weeks of the submission deadline.

Assessment Offences

The university's Assessment Offences Policy requires that you submit work that is entirely your own and reflects your own learning. It is important to:

- Ensure that you reference all sources used, using the given formatting guidelines.
- Avoid copying and pasting any work into this assessment, including your own previous assessments, work from other students or internet sources.
- Develop your own style, arguments and wording. Avoid copying and changing individual words but keeping essentially the same sentences and/or structures from other sources.
- Never give your work to others who may copy it.
- If you are doing an individual assessment, develop your own work and preparation. Do not allow anyone to make amendments to your work (including proof-readers, who may highlight issues but not edit the work).

When submitting your work, you will be required to confirm that the work is your own. Text-matching software and other methods are routinely used to check submissions against other submissions to the university and internet sources. Details of what constitutes plagiarism and how to avoid it can be found on UWE's Study Skills pages about avoiding plagiarism.

Marking Criteria for Project Proposal

	F	D	C	B	A
Project Proposal Relates to MO2, MO3 and MO4	More than three of the criteria for A are not fully met. More than three aspects of the proposal need to be paid more attention to.	Three of the criteria for A are not fully met. Three aspects of the proposal need to be paid more attention to.	Two of the criteria for A are not fully met. Two aspects of the proposal need to be paid more attention to.	One of the criteria for A is not fully met. One aspect of the proposal needs to be paid more attention to.	Application problem clearly defined and suitable for the project with the appropriate depth and scope. Aims and objectives clear. Dataset suitable and sufficient for the project. Proposed methodology suitable for solving the application problem. Proposed implementation has appropriate depth and scope. Poposed evaluation appropriate. No aspect of the proposal needs to be paid more attention to.

Marking Criteria for Final Submission

	0-39%	40-49%	50-69%	70-100%
Introduction (10%) Relates to MO2 and MO4	Identification of the problem without clear definition and the proposed approach without clear description, partial description of the aims and objectives, no or partial description of the results	Sufficient definition of the problem and description of the proposed approach, sufficient description of most of the aims, objectives and results	Good definition of the problem, good description of the proposed approach, good description of all the aims, objectives and results	Clear definition of the problem, excellent description of the proposed approach, excellent description of all the aims, objectives and results
Related Work (10%) Relates to MO2	Identification and discussion of little related work	Sufficient review of related work, satisfactory discussion of similarities and differences with some critical analysis	Good review of related work, good discussion of similarities and differences with good critical analysis linked to the project	Critical appraisal of related work, excellent discussion of similarities and differences with critical analysis related closely and clearly to the project
Data (10%) Relates to MO2 and MO3	Use of appropriate techniques to collect datasets or select existing datasets, treatment of data	Use and justification of appropriate techniques to collect datasets or select existing datasets, sufficient treatment of data	Good use and justification of appropriate techniques to collect datasets or select existing datasets, good treatment of data	Clear justification of appropriate techniques to collect datasets or select existing datasets; Excellent treatment of data
Methods (30%) Relates to MO3 and MO4	Selection and use of methods, basic level of technical breadth and depth, identification of relevant issues including ethical issues	Selection and use of methods, sufficient level of technical breadth and depth, sufficient discussion of relevant issues including ethical issues	Good selection and use of methods, good level of technical breadth and depth, good discussion of relevant issues including ethical issues	Excellent selection and use of methods demonstrating an understanding of alternative methods, high level of technical breadth and depth, excellent discussion of relevant issues including ethical issues
Experimental Evaluation (30%) Relates to MO3	Generation and analysis of experimental results	Generation and analysis of experimental results using justified methodology, appraisal and evaluation of results	Generation and analysis of experimental results using clearly justified methodology, good appraisal and evaluation of results	Excellent generation and clear analysis of experimental results using fully justified methodology, excellent appraisal and evaluation of results

Conclusion (5%) Relates to MO2 and MO4	Concise summary	Summary showing wider understanding	Good summary showing wider understanding	Demonstration of clear understanding and implications of results
Writing and Formatting (5%) Relates to MO2, MO3 and MO4	Use of template and basic writing skills	Satisfactory structure, some graphical/tabular illustrations.	Coherent structure, good use of graphical/tabular illustrations.	Well structured, high-level of clarity, clear presentation and illustrations